



STM Exercise 9

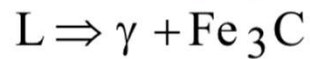
w

Phase Diagrams
(Fe-C)

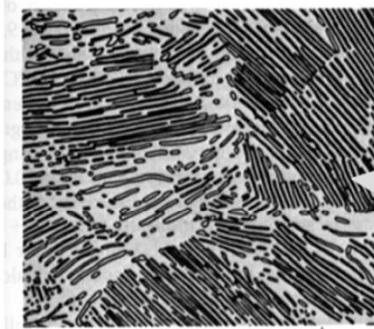
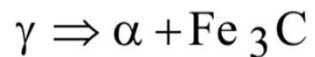
IRON-CARBON (Fe-C) PHASE DIAGRAM

- 2 important points

-Eutectic (A):

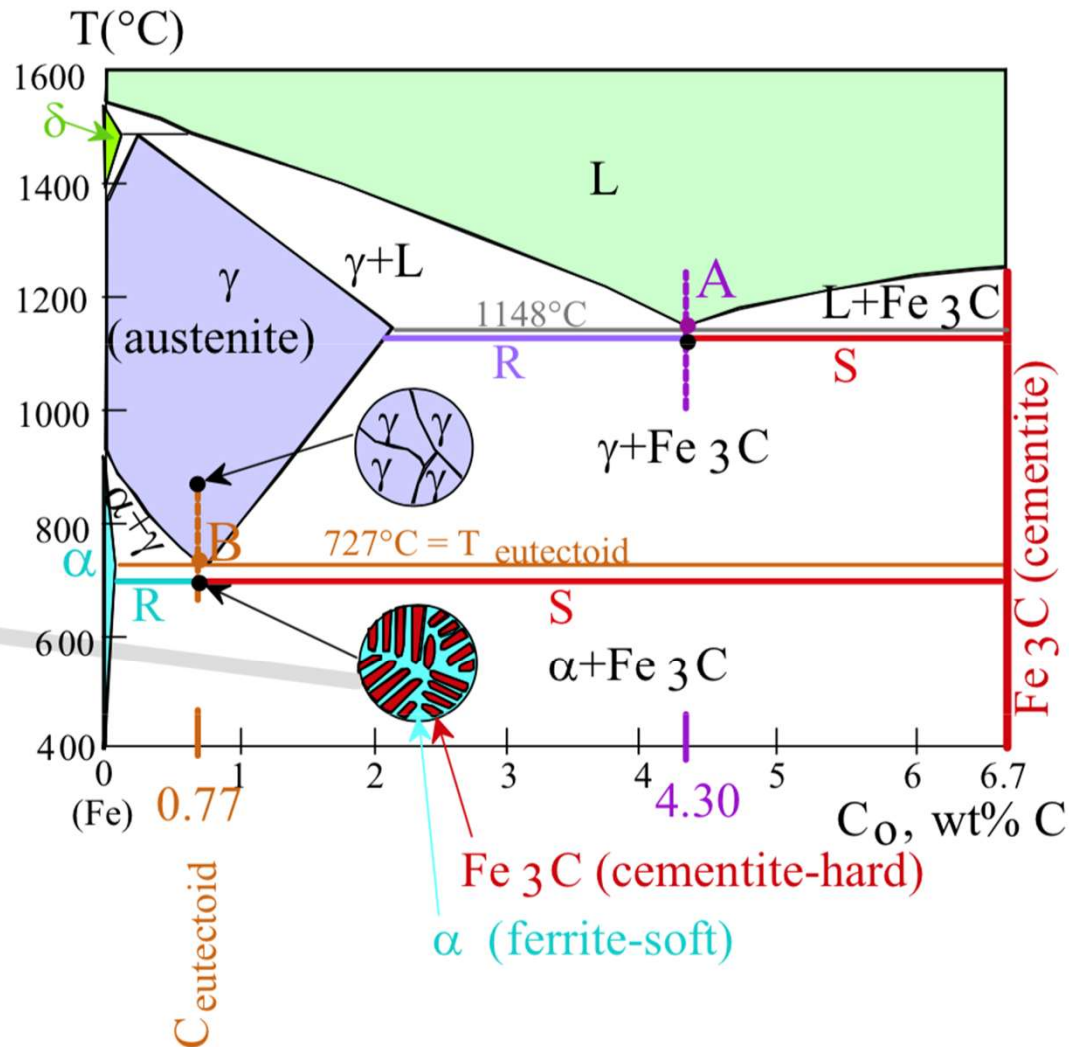


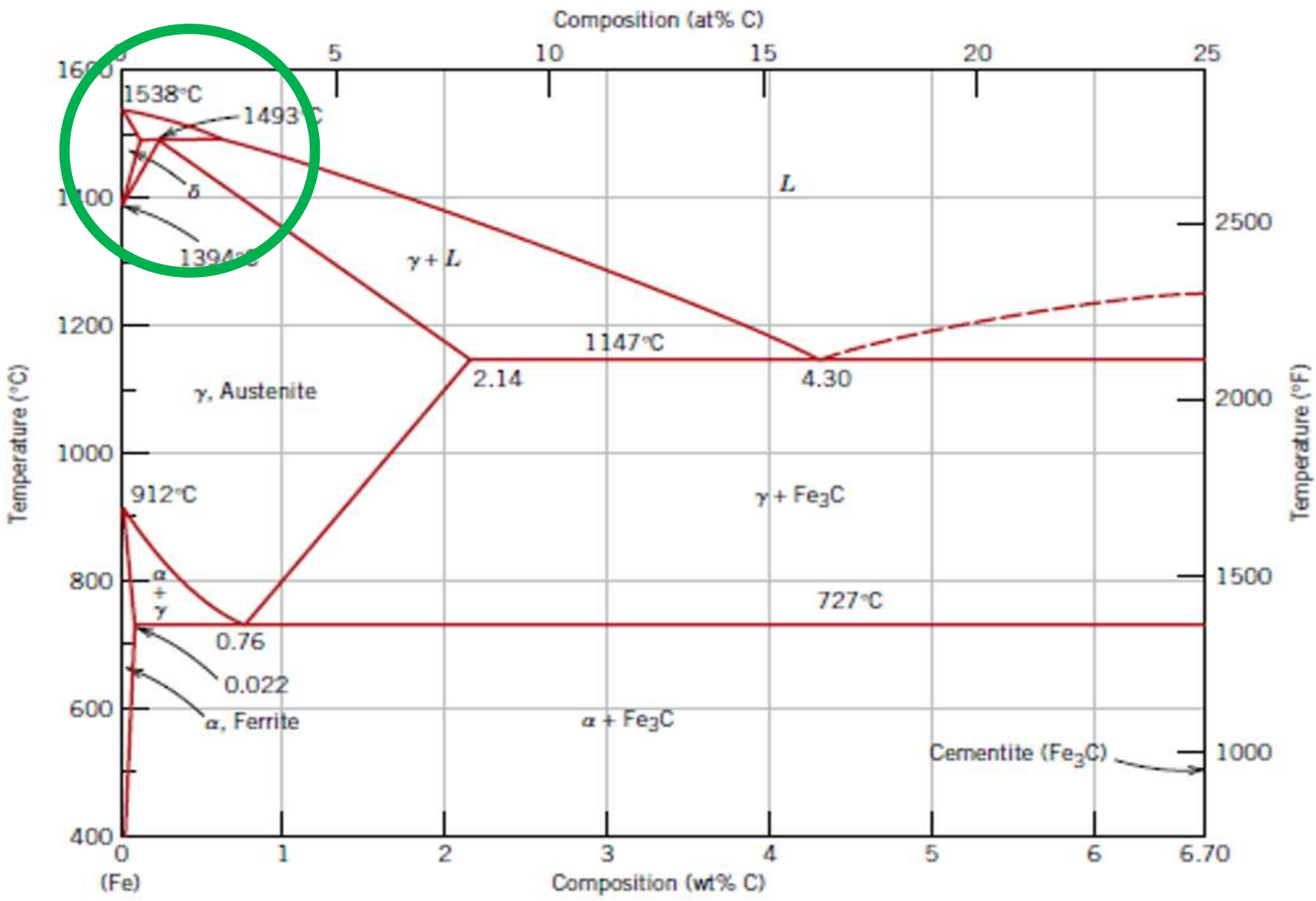
-Eutectoid (B):

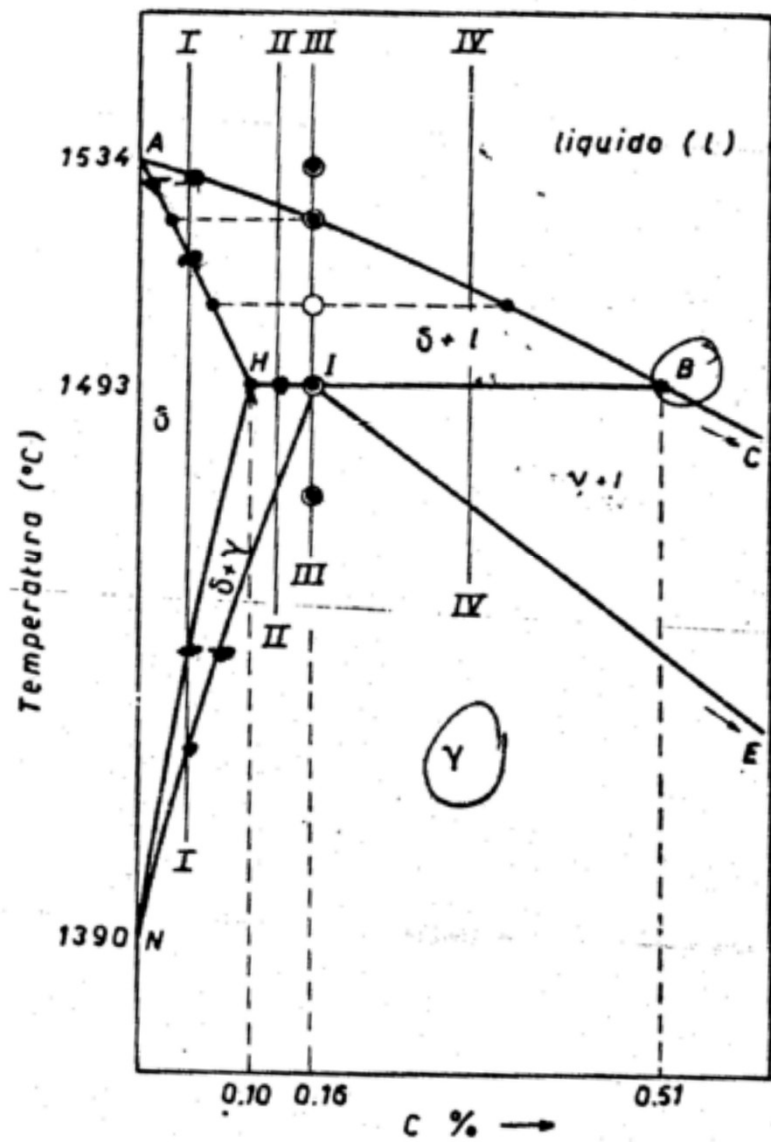


120 μm

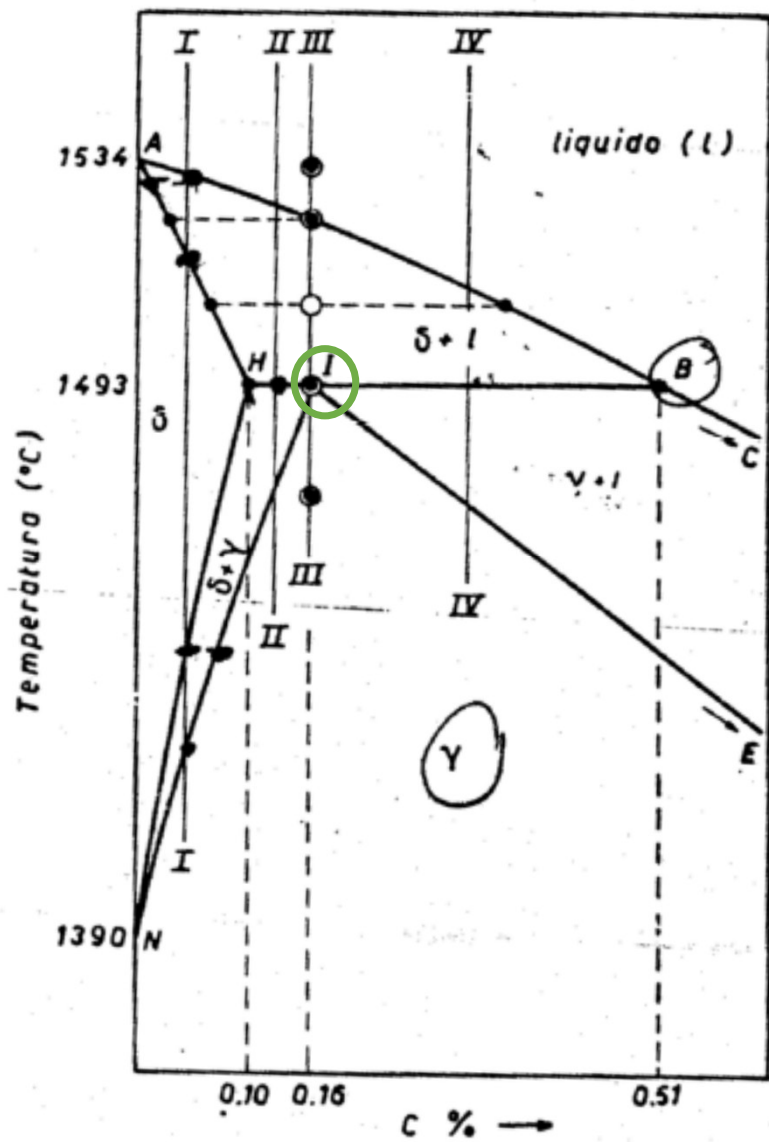
Result: Pearlite = alternating layers of α and Fe_3C phases.





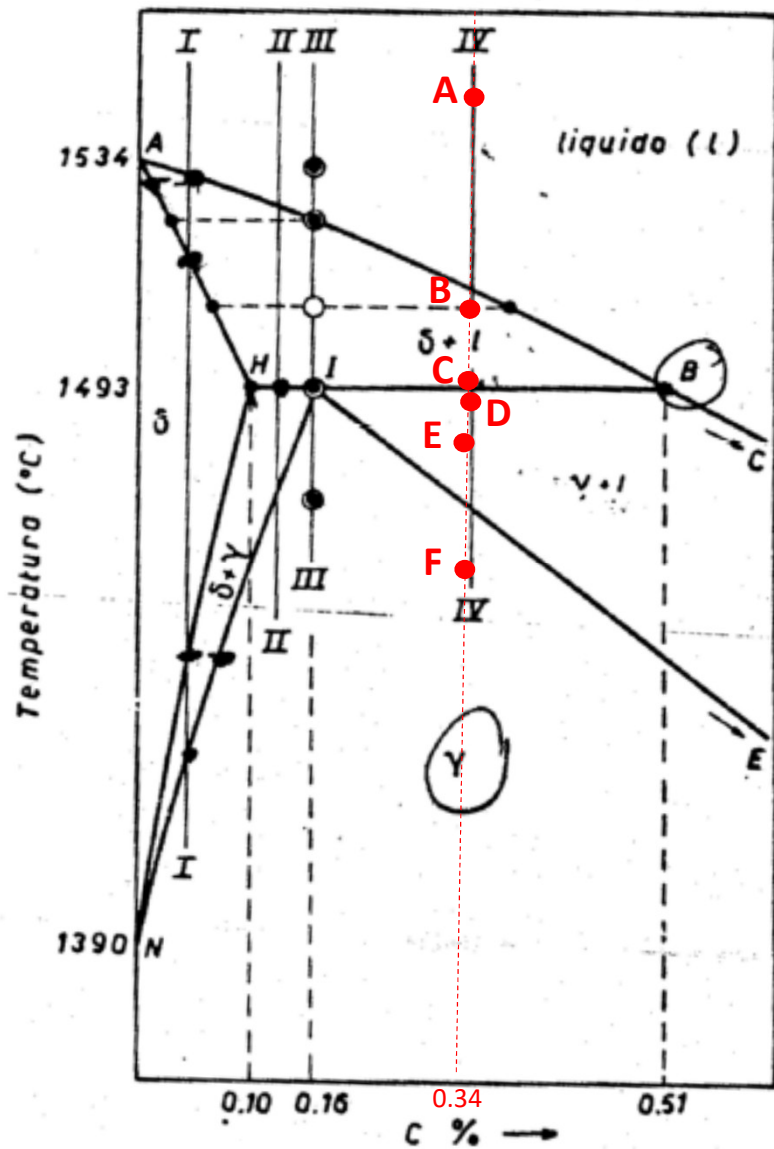


Exercise 1: which transformation occurs in point I?

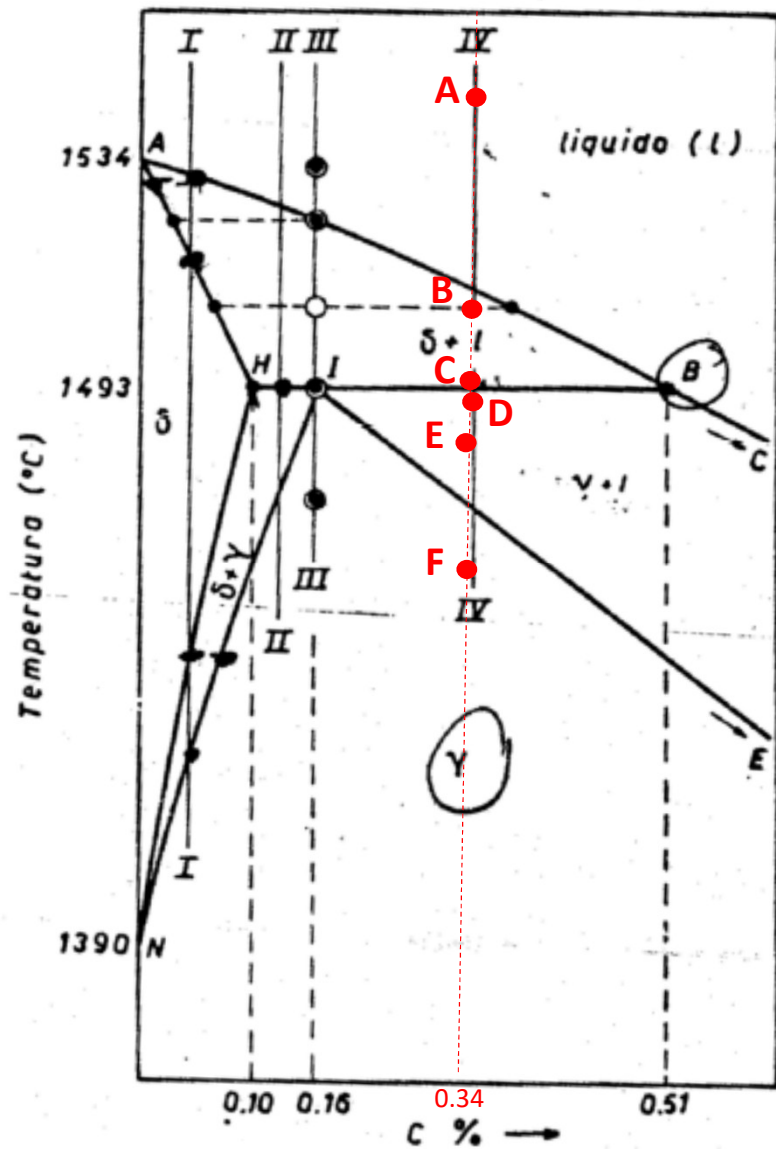


Exercise 1: which transformation occurs in point I?

Point I is a **PERITECTIC** transformation
 $\delta + L \leftrightarrow \gamma$



Exercise 2: Determine the phase/s present their compositions and relative amount in points A, B, C (at the beginning of the peritectic transformation), D(at the end of the peritectic transformation), E and F for a steel with 0.34%C.

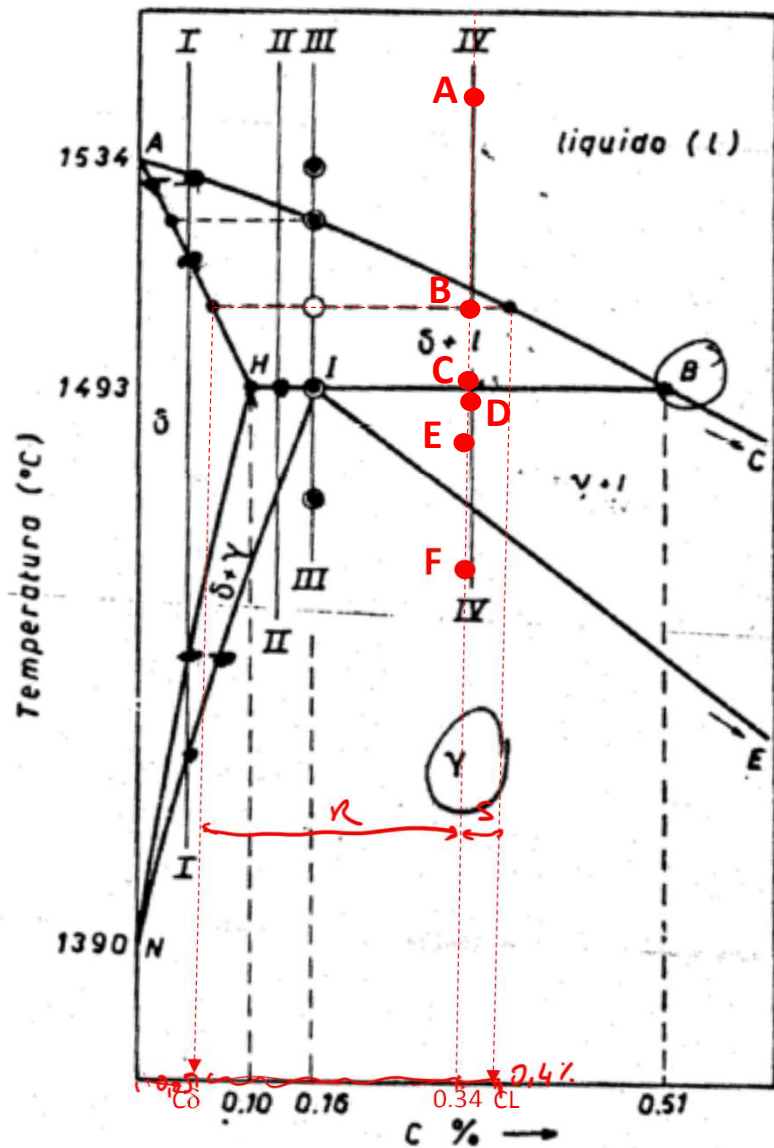


Point A:

1 phase liquid

Composition 0.34%C

Relative amount 100% liquid



Point B:

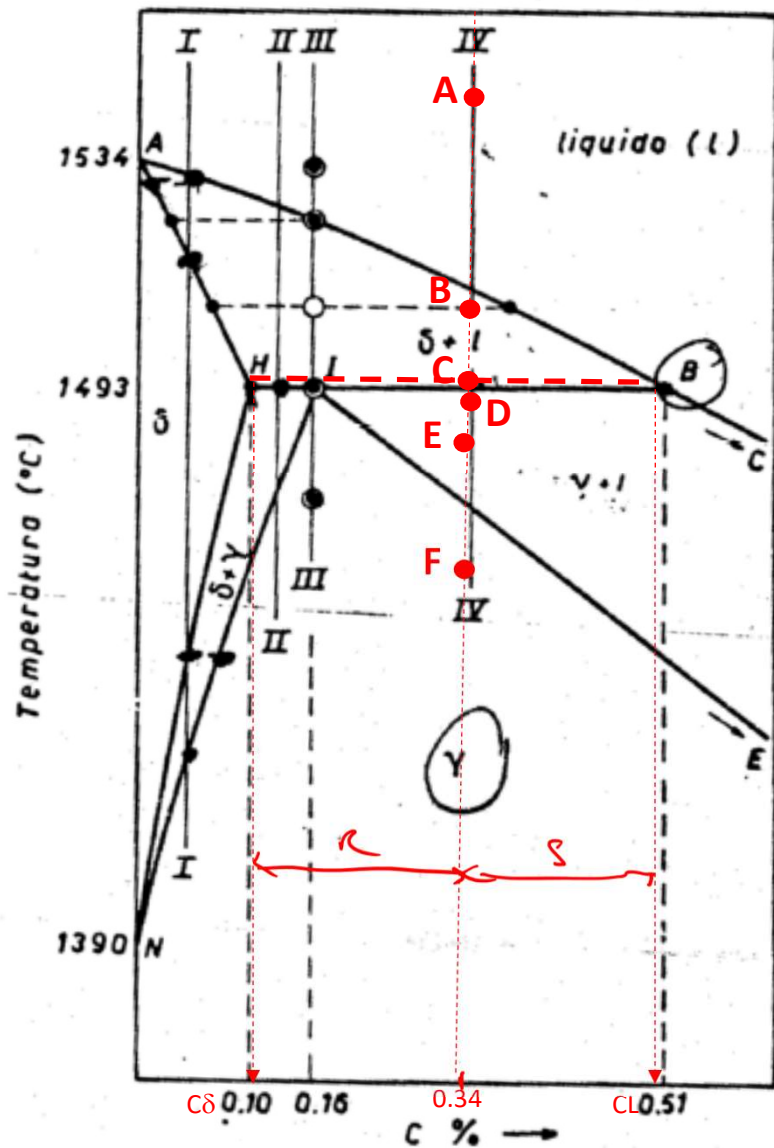
2 phases δ ferrite (solid) and liquid

Compositions (HORIZONTAL RULE)

- Composition of δ ferrite : $C_{\delta}=0.05\%C$
- Composition of liquid: $C_L=0.4\%C$

Relative amounts (LEVER RULE)

- $\% \delta \text{ ferrite} = \frac{(0.4-0.34)}{(0.4-0.05)} \times 100 = 17\%$
- $\% \text{ Liquid} = \frac{(0.34-0.05)}{(0.4-0.05)} \times 100 = 83\%$



Point C:

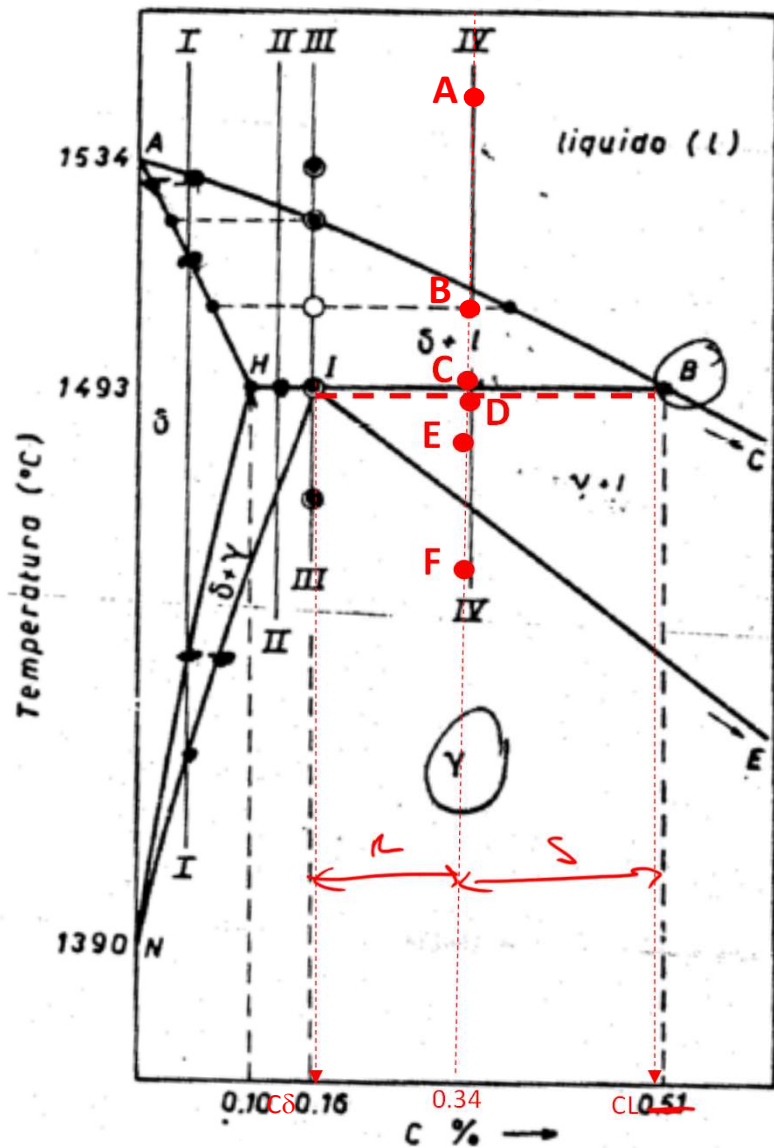
2 phases δ ferrite (solid) and liquid

Compositions (HORIZONTAL RULE)

- Composition of δ ferrite : $C_{\delta}=0.1\%C$
- Composition of liquid: $C_L=0.51\%C$

Relative amounts (LEVER RULE)

- $\% \delta \text{ ferrite} = \frac{(0.51-0.34)}{(0.51-0.1)} \times 100 = 41.5\%$
- $\% \text{ Liquid} = \frac{(0.34-0.1)}{(0.51-0.1)} \times 100 = 58.5\%$



Point D:

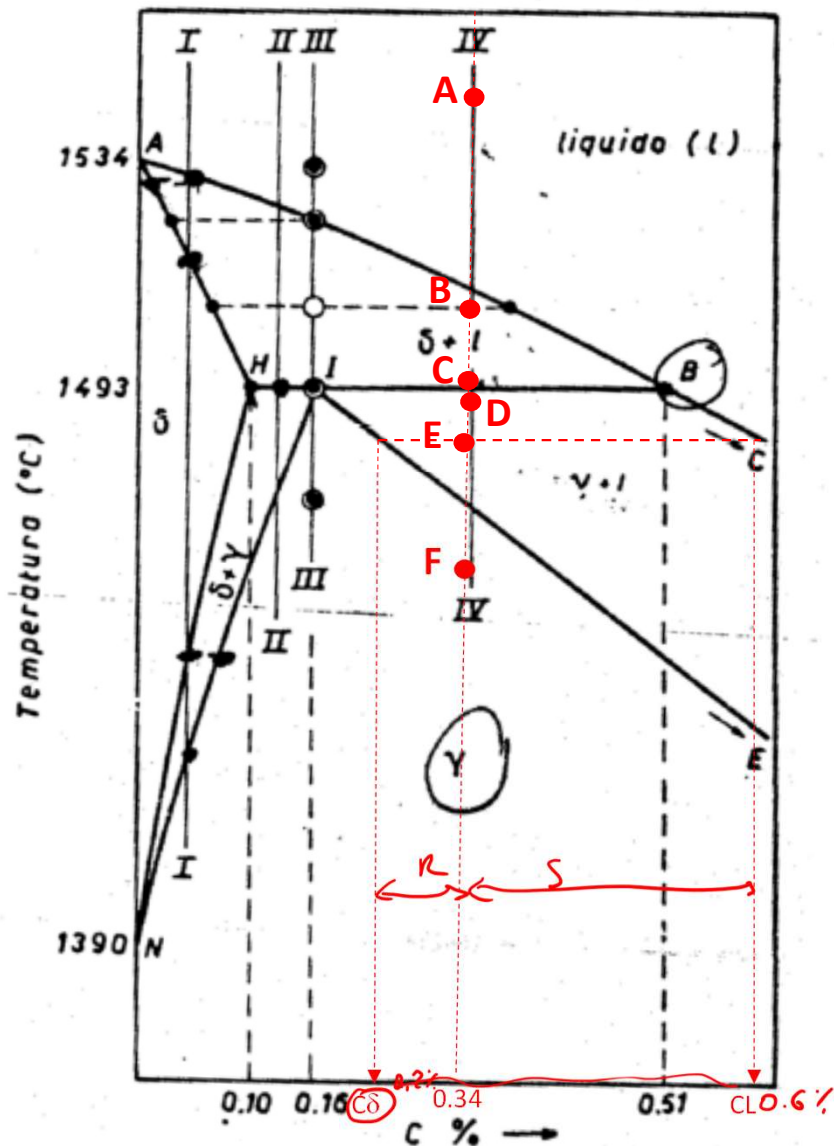
2 phases γ austenite (solid) and liquid

Compositions (HORIZONTAL RULE)

- Composition of γ austenite : $C_{\delta} = 0.16\%C$
- Composition of liquid: $C_L = 0.51\%C$

Relative amounts (LEVER RULE)

- % γ austenite = $\frac{(0.51 - 0.34)}{(0.51 - 0.16)} \times 100 = 48.5\%$
- % Liquid = $\frac{(0.34 - 0.16)}{(0.51 - 0.16)} \times 100 = 51.5\%$



Point E:

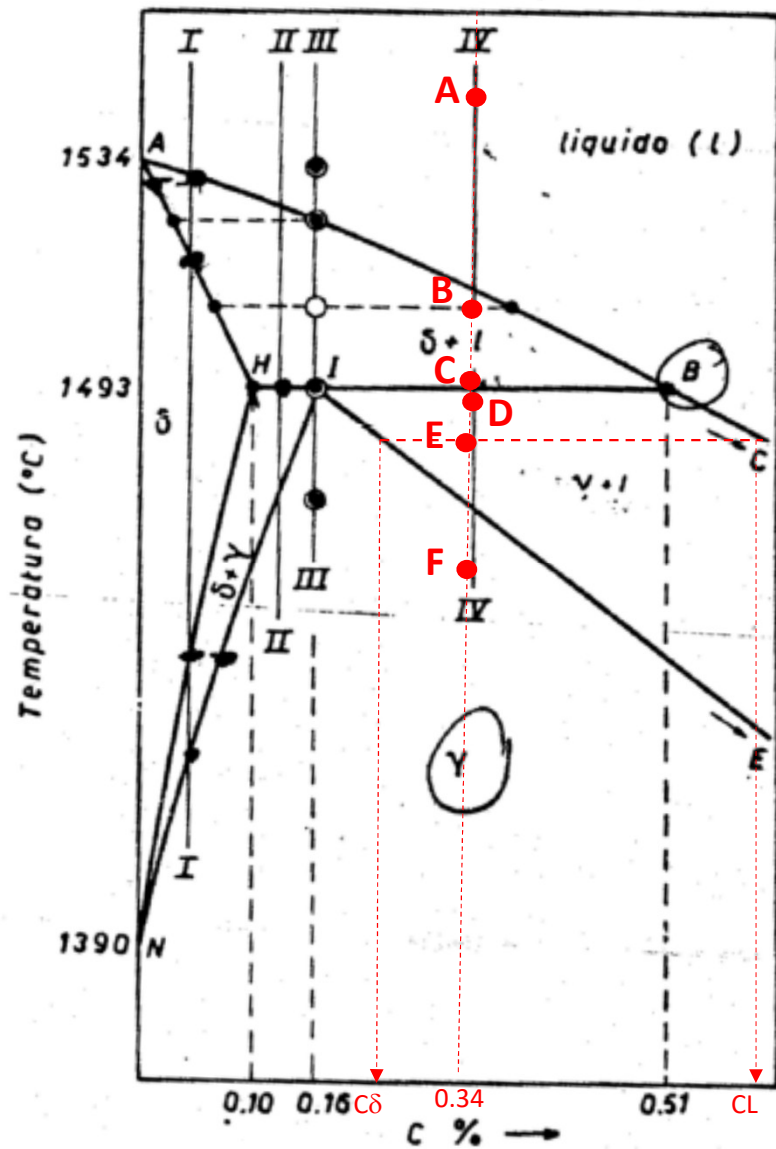
2 phases γ austenite (solid) and liquid

Compositions (HORIZONTAL RULE)

- Composition of γ austenite : $C_{\delta}=0.2\%C$
- Composition of liquid: $C_L=0.6\%C$

Relative amounts (LEVER RULE)

- $\% \gamma \text{ austenite} = \frac{(0.6 - 0.34)}{(0.6 - 0.2)} * 100 = 65\%$
- $\% \text{Liquid} = \frac{(0.34 - 0.2)}{(0.6 - 0.2)} * 100 = 35\%$

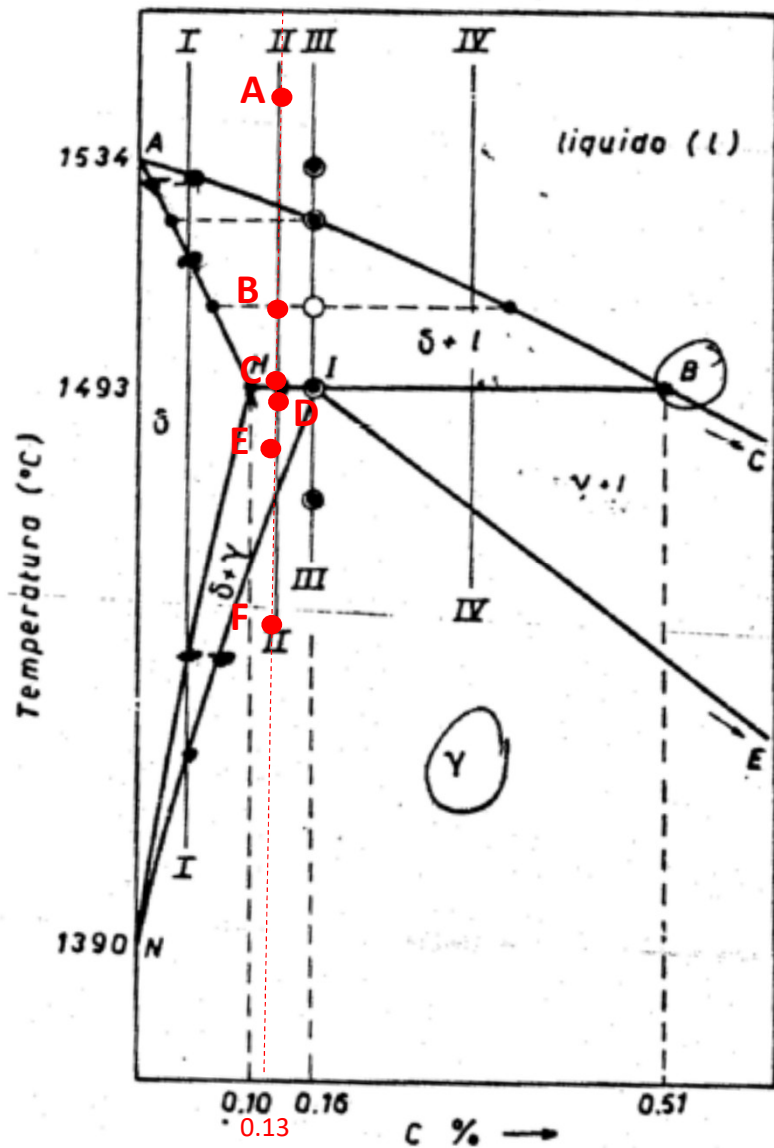


Point F:

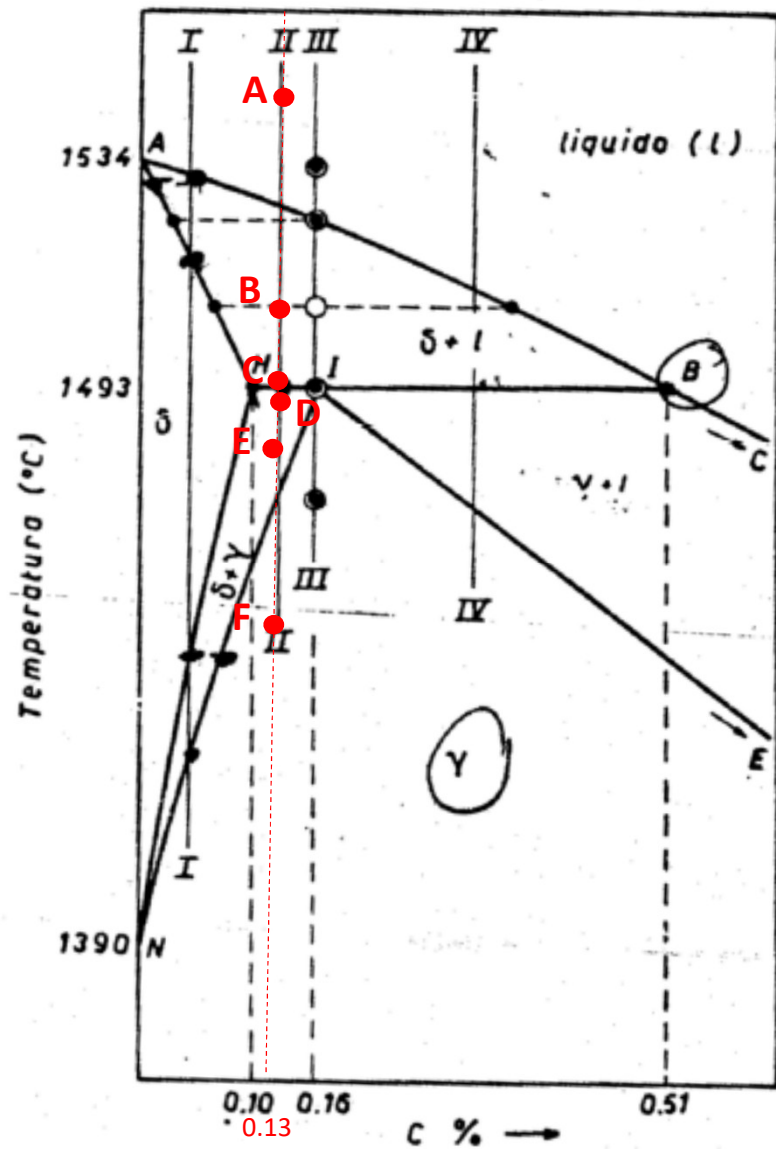
1 phase γ austenite (solid)

Composition: 0.34%C

Relative amount 100% solid γ austenite



Exercise 3: Determine the phase/s present their compositions and relative amount in points A, B, C (at the beginning of the peritectic transformation), D(at the end of the peritectic transformation), E and F for a steel with 0.13%C.

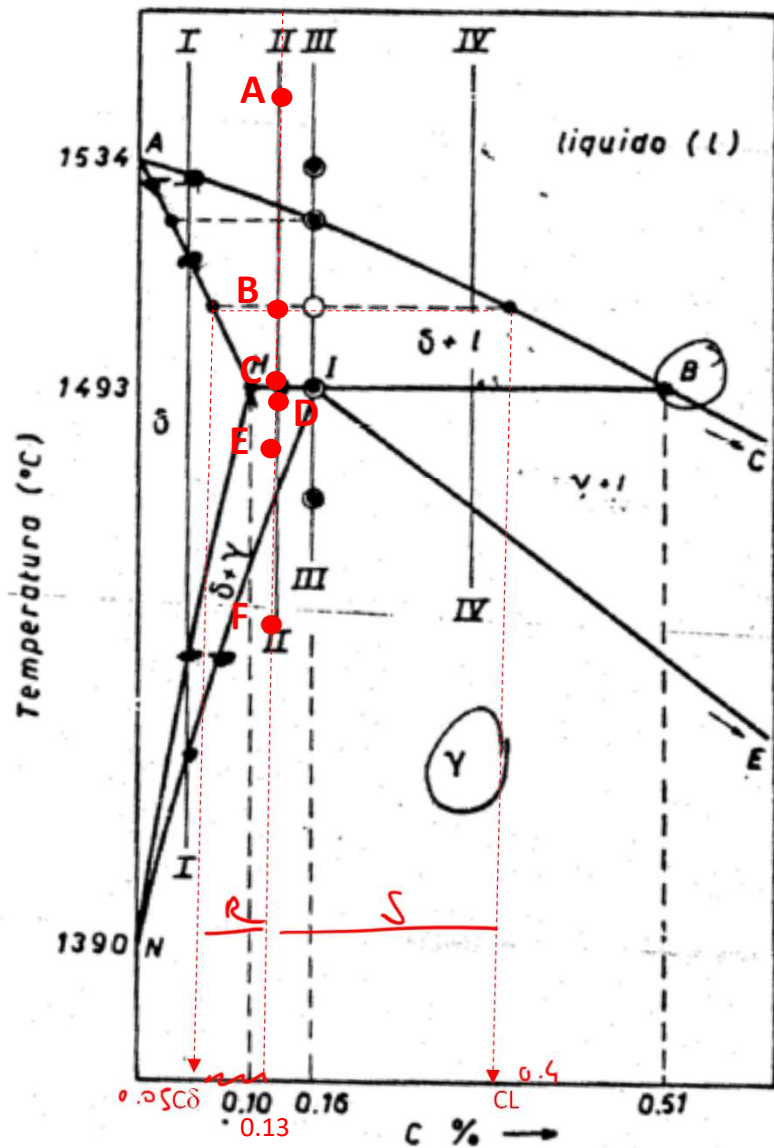


Point A:

1 phase liquid

Composition 0.13%C

Relative amount 100% liquid



Point B:

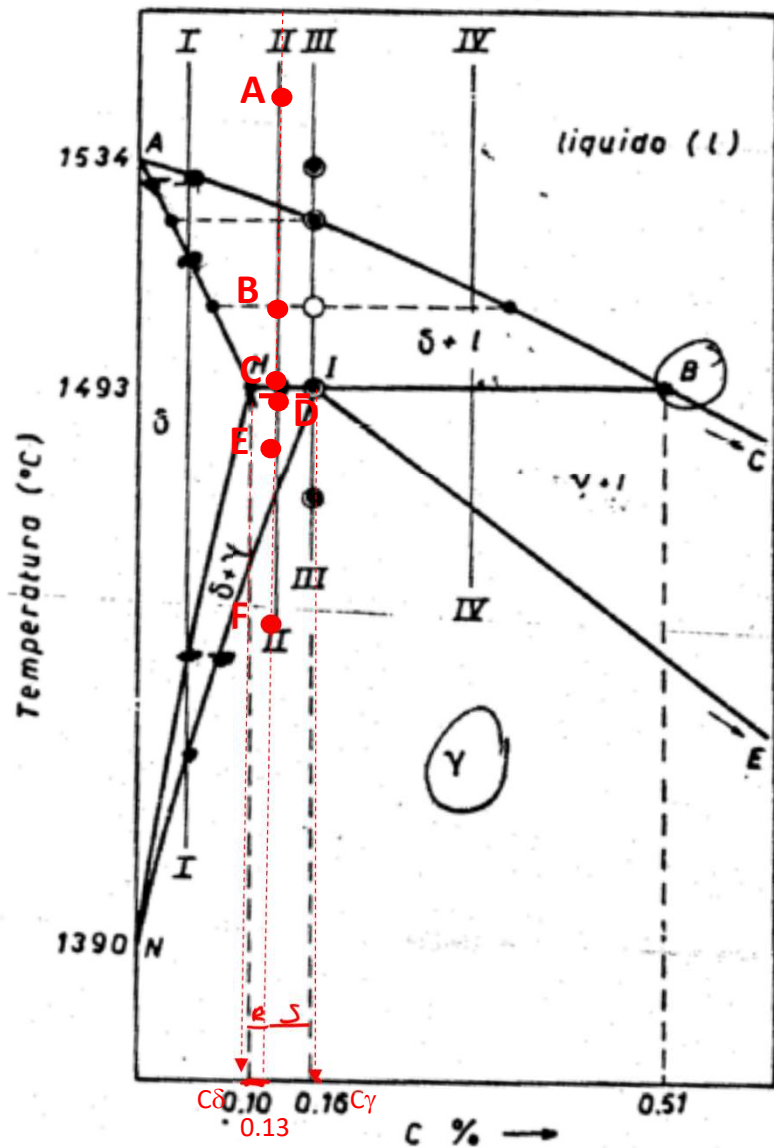
2 phases δ ferrite (solid) and liquid

Compositions (HORIZONTAL RULE)

- Composition of δ ferrite : $C_{\delta}=0.05\%C$
- Composition of liquid: $CL=0.4\%C$

Relative amounts (LEVER RULE)

- $\% \delta \text{ ferrite} = \frac{(0.4 - 0.13)}{(0.4 - 0.05)} * 100 = 77\%$
- $\% \text{ Liquid} = \frac{(0.13 - 0.05)}{(0.4 - 0.05)} * 100 = 23\%$



Point D:

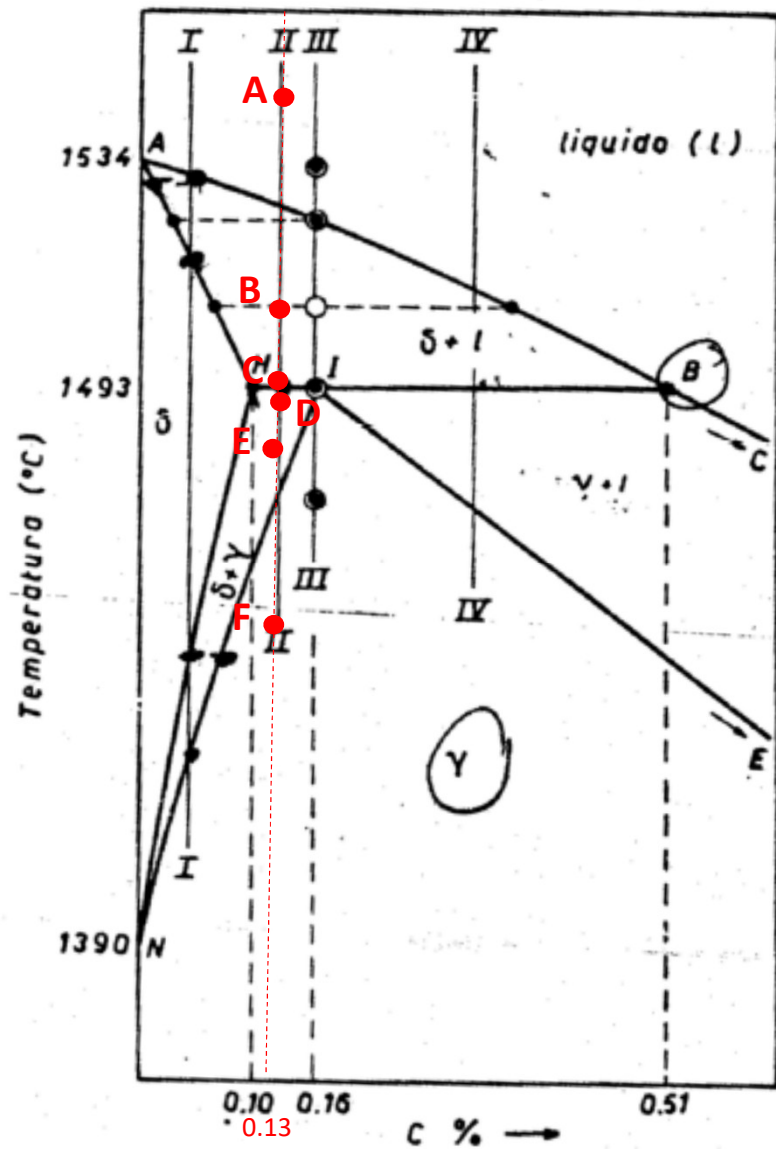
2 phases δ ferrite (solid) and γ austenite (solid)

Compositions (HORIZONTAL RULE)

- Composition of δ ferrite : $C_{\delta}=0.1\%C$
- Composition of γ austenite (solid): $C_{\gamma}=0.16\%C$

Relative amounts (LEVER RULE)

- $\% \delta \text{ ferrite} = \frac{(0.16-0.13)}{(0.16-0.1)} \times 100 = 50\%$
(Handwritten: S above 0.16, S+L above 0.1)
- $\% \gamma \text{ austenite} = \frac{(0.13-0.1)}{(0.16-0.1)} \times 100 = 50\%$
(Handwritten: L below 0.13, S+L below 0.1)



Point F:

1 phase γ austenite (solid)

Composition: 0.13%C

Relative amount 100% solid γ austenite